Data analytics for non life insurance (II)
GLM

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MESIO

Generalized Linear Models for the cost of claims

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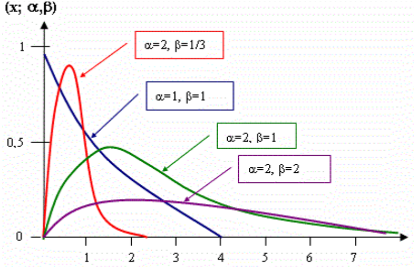
Gamma regression model

- The Gamma distribution has two parameters: shape, $\boldsymbol{\alpha}$, and scale, $\boldsymbol{\beta}$.

- The density function is:

$$f(y;\alpha,\beta) = \frac{\beta^{\alpha}\ y^{\alpha-1}\ e^{-\beta y}}{\Gamma(\alpha)} \quad \alpha,\beta > 0\ ,\ \forall y \in \mathbb{R}_{+}$$

- The mean of the distribution is $E[Y] = \alpha/\beta$ and the variance is $Var[Y] = \alpha/\beta^2$.



Gamma regression model

- Changing the parameters, $\mu = \alpha/\beta$ y $\alpha = 1/\phi = \phi^{-1}$, we have:

$$f(y;\mu,\phi) = \frac{1}{\Gamma\left(\frac{1}{\phi}\right)} \left(\frac{1}{\mu\phi}\right)^{\frac{1}{\phi}} y^{\frac{1}{\phi}-1}\, e^{-y/\mu\phi}$$

- And changing the order of the parameters:

$$f(y;\mu,\phi) = \exp\left[-\ln\Gamma\left(\frac{1}{\phi}\right) - \frac{1}{\phi}\ln\phi - \frac{1}{\phi}\ln\mu + \left(\frac{1}{\phi}-1\right)\ln y - \frac{y}{\phi\mu}\right] =$$

$$= \exp\left[\frac{y\left(\frac{-1}{\mu}\right) - \ln\mu}{\phi} + \left(\frac{1}{\phi}-1\right)\ln y - \ln\Gamma\left(\frac{1}{\phi}\right) - \frac{1}{\phi}\ln\phi\right]$$

Gamma regression model

The former expression belongs to the exponential family:

$$f(y;\mu,\phi) = \exp\left[\frac{y\left(\frac{-1}{\mu}\right) - \ln\mu}{\phi} + \left(\frac{1}{\phi}-1\right)\ln y - \ln\Gamma\left(\frac{1}{\phi}\right) - \frac{1}{\phi}\ln\phi\right]$$

if we consider that:

- $\theta = -1/\mu$ (inverse canonic link),
- $b(\theta) = \ln\mu$,
- $c(y,\phi) = \left(\frac{1}{\phi}-1\right)\ln y - \ln\Gamma\left(\frac{1}{\phi}\right) - \frac{1}{\phi}\ln\phi$.

As $\theta = -1/\mu$ y $\mu = \alpha/\beta$ is positive $(\alpha, \beta > 0)$, then $\theta < 0$.

If we substitute $-1/\theta = \mu$ then $b(\theta) = \ln(-1/\theta) = -\ln(-\theta)$.

The mean of the distribution is: $b'(\theta) = \frac{d}{d\theta}(-\ln(-\theta)) = \frac{-1}{\theta} = \mu$

and the variance: $\phi b''(\theta) = \phi\frac{1}{\theta^2} = \phi\mu^2$, and the variance function $V(\mu) = \mu^2$

Gamma regression model

- In the case of the Gamma GLM we can use the following links:
  - **Log link:** The most frequenly used

    $g(\mu_i) = \ln(\mu_i) \quad \Rightarrow \quad \mu_i = \exp(x_i'\beta) \quad \Rightarrow \quad \log(\mu_i) = x_i'\beta$

    (a graph of $\ln Y_i$ versus x_i should be linear)
  - **Inverse link:**

    $g(\mu_i) = \frac{1}{\mu_i} \quad \Rightarrow \quad \mu_i = \frac{1}{x_i'\beta} \quad \Rightarrow \quad \frac{1}{\mu_i} = x_i'\beta$ (canonical link: $\frac{-1}{\mu}$)

    (a graph of $\mathbf{1}/Y_i$ versus x_i should be linear)
  - **Identity link:**

    $g(\mu_i) = \mu_i$

Examples with R: Gamma

With link=log $\Rightarrow \mathrm{E}(\mathrm{Y} \mid \mathrm{X}) = \exp[\mathrm{X}'\beta]$

```
> car1=car[car$claimcst0>0,]
> # Gamma Model
> gama= glm(claimcst0~gender+AgeCat,family=Gamma(link="log"),data=car1)
> summary(gama)
Call:
glm(formula = claimcst0 ~ gender + AgeCat, family = Gamma(link = "log"),data = car1)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.43592    0.09354  79.498  < 2e-16 ***
genderM      0.17508    0.05122   3.418 0.000635 ***
AgeCat1      0.34806    0.11854   2.936 0.003338 **
AgeCat2      0.14865    0.10618   1.400 0.161566
AgeCat3      0.05603    0.10383   0.540 0.589447
AgeCat4      0.05933    0.10384   0.571 0.567802
AgeCat5     -0.06051    0.11358  -0.533 0.594250
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for Gamma family taken to be 2.952696)
    Null deviance: 7379.9  on 4623  degrees of freedom
Residual deviance: 7283.4  on 4617  degrees of freedom
AIC: 79364
Number of Fisher Scoring iterations: 6
> library("MASS")
> gamma.shape(gama)
Alpha: 0.75889423
SE:    0.01354924
```

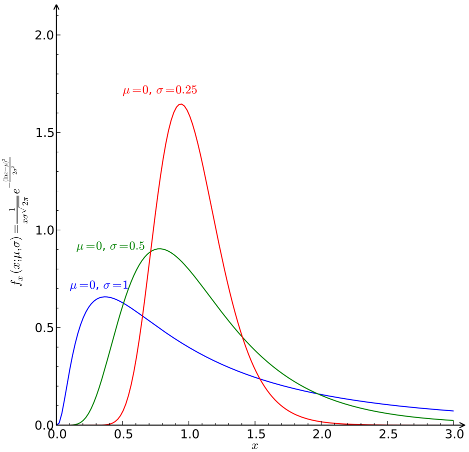
Log-normal regression model

- The Log-normal model is also suitable when the variable has positive skewness.
- A random variable is Log-normally distributed if its logaritmic transformation has a Normal distribution.
- Therefore, if $\ln(c) \sim N(\mu_c, \sigma_c^2)$, then, by doing a variable transformation we have that:

$$f(c) = \frac{1}{c\sqrt{2\pi\sigma_c^2}} \exp\left\{-\frac{1}{2}\left[\frac{\ln(c)-\mu_c}{\sigma_c}\right]^2\right\} \quad , \forall c > 0$$

and the mean is $E[c] = e^{\left[\mu_c + \frac{1}{2}\sigma_c^2\right]}$ and the variance is $Var[c] = e^{\left[2\mu_c + \sigma_c^2\right]}(e^{\sigma_c^2 - 1})$.

Log-normal regression model



- The bigger σ_c^2 is, the bigger the righ skewness is.

Examples with R: Log-normal

With $\mathrm{E}(\ln \mathrm{Y} \mid \mathrm{X}) = \mathrm{X}'\beta$

```
> # Log-normal Model
> regln=lm(log(claimcst0)~gender+AgeCat,data=car1)
> summary(regln)

Call:
lm(formula = log(claimcst0) ~ gender + AgeCat, data = car1)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.743075   0.064620 104.350  < 2e-16 ***
genderM      0.088581   0.035384   2.503  0.01233 *
AgeCat1      0.217764   0.081893   2.659  0.00786 **
AgeCat2      0.039604   0.073353   0.540  0.58928
AgeCat3      0.021029   0.071729   0.293  0.76940
AgeCat4      0.003213   0.071738   0.045  0.96428
AgeCat5     -0.060383   0.078467  -0.770  0.44162
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.187 on 4617 degrees of freedom
Multiple R-squared:  0.004963,        Adjusted R-squared:  0.003669
F-statistic: 3.838 on 6 and 4617 DF,  p-value: 0.0008036
```

Inverse Gaussian regression model

- The Inverse Gaussian model, with density function:

$$f(y) = \left[\frac{\lambda}{2\pi y^3}\right]^{1/2} \exp\left(\frac{-\lambda(y-\mu)^2}{2\mu^2 y}\right) ; \ \mu, \lambda > 0 \ \forall y \in \mathbb{R}_+$$

also belongs to the exponential family, with:

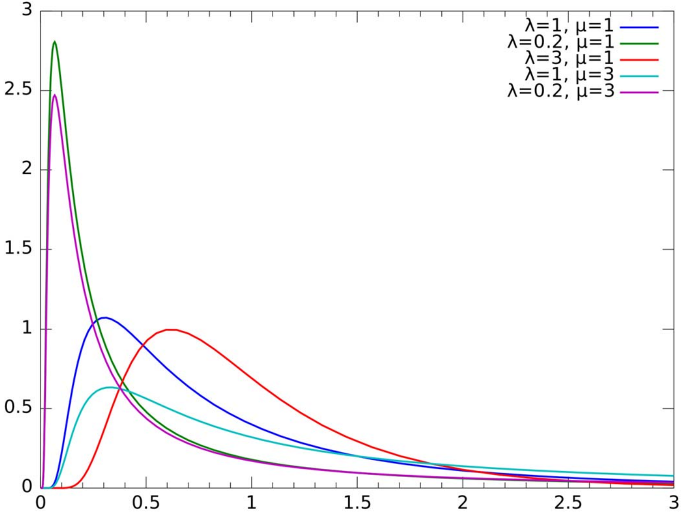
- $\theta = -1/2\mu^2$ y $\phi = 1/\lambda$,
- $b(\theta) = -\sqrt{-2\theta}$,
- $c(y, \phi) = -\frac{1}{2}\left\{\ln(\frac{2\pi y^3}{\lambda}) + \frac{\lambda}{y}\right\}$.

The mean of the distribution will be: $E(Y) = \mu = b'(\theta) = (-2\theta)^{-1/2}$,

and the variance: $Var(Y) = \phi b''(\theta) = \phi(-2\theta)^{-3/2} = \phi\mu^3 = \mu^3/\lambda$,

and the variance function is $V(\mu) = \mu^3$

Inverse Gaussian regression Model



Examples with R: Inverse Gaussian

With un link=log $\Rightarrow \mathrm{E}(\mathrm{Y} \mid \mathrm{X}) = \exp[\mathrm{X}'\beta]$

```
> # Inverse Gaussian Model
> invgaus=glm(claimcst0~gender+AgeCat,data=car1,family=inverse.gaussian(link="log"),
                                                               start=coefficients(gama))
> summary(invgaus)

Call:
glm(formula = claimcst0 ~ gender + AgeCat, family = inverse.gaussian(link = "log"),
    data = car1, start = coefficients(gama))

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.43293    0.08942  83.124  < 2e-16 ***
genderM      0.16642    0.05176   3.216  0.00131 **
AgeCat1      0.34422    0.12280   2.803  0.00508 **
AgeCat2      0.15393    0.10390   1.482  0.13853
AgeCat3      0.06603    0.10009   0.660  0.50948
AgeCat4      0.06560    0.10017   0.655  0.51256
AgeCat5     -0.04718    0.10782  -0.438  0.66168
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for inverse.gaussian family taken to be 0.001480413)

    Null deviance: 6.4422  on 4623  degrees of freedom
Residual deviance: 6.3972  on 4617  degrees of freedom
AIC: 77167
```

Examples with SAS:

```
data car1; set lib.car;
 if claimcst0 >0;
 lcost=log(claimcst0);
run;

/* Gamma with link=log */
proc genmod data=car1;
 class gender ( Ref = 'F' param = ref ) agecat ( Ref = '6' param=ref );
 model claimcst0 = gender agecat / dist=GAMMA link=log;
run;

/* Inverse Gaussian with link=log */
proc genmod data=car1;
  class gender (Ref='F' param=ref) agecat (Ref='6' param=ref);
  model claimcst0 = gender agecat / dist=IGAUSSIAN link=log;
 run;

/* Log-normal */
proc genmod data=car1;
  class gender (Ref='F' param=ref) agecat (Ref='6' param=ref);
  model lcost = gender agecat / dist=NORMAL link=IDENTITY;
run;
```

Examples with SAS

With link=log $\Rightarrow \mathrm{E}(\mathrm{Y} \mid \mathrm{X}) = \exp[\mathrm{X}'\beta]$

Procedimiento GENMOD

Distribución	Gamma
Función de vínculo	Log
Variable dependiente	claimcst0

Número de observaciones usadas 4624

Criterios para valorar la bondad de ajuste

Criterio	DF	Valor	Valor/DF
Desviación	4617	7283.4233	1.5775
Desviación escalada	4617	5527.3475	1.1972
Chi-cuadrado de Pearson	4617	13632.5136	2.9527
Pearson X2 escalado	4617	10345.6352	2.2408
Verosimilitud log		-39626.5315	
Verosimilitud log completa		-39626.5315	
AIC (mejor más pequeño)		79269.0630	
AICC (mejor más pequeño)		79269.0942	
BIC (mejor más pequeño)		79320.5751	

Algoritmo convergido.

Análisis de estimadores de parámetros de máxima verosimilitud

Parámetro		DF	Estimador	Error estándar	Wald 95% Límites de confianza		Chi-cuadrado de Wald	Pr > ChiSq
Intercept		1	7.4359	0.0629	7.3126	7.5593	13959.4	<.0001
gender	M	1	0.1751	0.0343	0.1078	0.2424	25.99	<.0001
agecat	1	1	0.3481	0.0792	0.1929	0.5032	19.33	<.0001
agecat	2	1	0.1487	0.0710	0.0095	0.2878	4.39	0.0362
agecat	3	1	0.0560	0.0695	-0.0803	0.1923	0.65	0.4203
agecat	4	1	0.0593	0.0694	-0.0768	0.1955	0.73	0.3929
agecat	5	1	-0.0605	0.0760	-0.2094	0.0884	0.63	0.4258
Escal		1	0.7589	0.0136	0.7328	0.7859		

NOTE: La verosimilitud máxima ha estimado el parámetro de escala.

Examples with SAS: Inverse Gaussian

With link=log $\Rightarrow \mathrm{E}(\mathrm{Y} \mid \mathrm{X}) = \exp[\mathrm{X}'\beta]$

Procedimiento GENMOD

Distribución	Inverse Gaussian
Función de vínculo	Log
Variable dependiente	claimcst0

Número de observaciones usadas 4624

Criterios para valorar la bondad de ajuste

Criterio	DF	Valor	Valor/DF
Desviación	4617	6.3972	0.0014
Desviación escalada	4617	4624.0000	1.0015
Chi-cuadrado de Pearson	4617	6.8351	0.0015
Pearson X2 escalado	4617	4940.5163	1.0701
Verosimilitud log		-38575.6437	
Verosimilitud log completa		-38575.6437	
AIC (mejor más pequeño)		77167.2873	
AICC (mejor más pequeño)		77167.3185	
BIC (mejor más pequeño)		77218.7994	

Algoritmo convergido.

Análisis de estimadores de parámetros de máxima verosimilitud

Parámetro		DF	Estimador	Error estándar	Wald 95% Límites de confianza		Chi-cuadrado de Wald	Pr > ChiSq
Intercept		1	7.4329	0.0879	7.2607	7.6052	7154.05	<.0001
gender	M	1	0.1664	0.0506	0.0673	0.2656	10.83	0.0010
agecat	1	1	0.3442	0.1187	0.1116	0.5768	8.41	0.0037
agecat	2	1	0.1539	0.1007	-0.0434	0.3512	2.34	0.1262
agecat	3	1	0.0660	0.0974	-0.1250	0.2570	0.46	0.4981
agecat	4	1	0.0656	0.0971	-0.1248	0.2560	0.46	0.4995
agecat	5	1	-0.0472	0.1047	-0.2524	0.1580	0.20	0.6521
Escal		1	0.0372	0.0004	0.0364	0.0380		

NOTE: La verosimilitud máxima ha estimado el parámetro de escala.

Examples with SAS: Log-normal

With $\mathrm{E}(\ln \mathrm{Y} \mid \mathrm{X}) = \mathrm{X}'\beta$

Procedimiento GENMOD

Distribución	Normal
Función de vínculo	Identity
Variable dependiente	lcost

Número de observaciones usadas 4624

Criterios para valorar la bondad de ajuste

Criterio	DF	Valor	Valor/DF
Desviación	4617	6506.5685	1.4093
Desviación escalada	4617	4624.0000	1.0015
Chi-cuadrado de Pearson	4617	6506.5685	1.4093
Pearson X2 escalado	4617	4624.0000	1.0015
Verosimilitud log		-7350.8402	
Verosimilitud log completa		-7350.8402	
AIC (mejor más pequeño)		14717.6803	
AICC (mejor más pequeño)		14717.7116	
BIC (mejor más pequeño)		14769.1925	

Algoritmo convergido.

Análisis de estimadores de parámetros de máxima verosimilitud

Parámetro		DF	Estimador	Error estándar	Wald 95% Límites de confianza		Chi-cuadrado de Wald	Pr > ChiSq
Intercept		1	6.7431	0.0646	6.6165	6.8696	10905.4	<.0001
gender	M	1	0.0886	0.0354	0.0193	0.1579	6.28	0.0122
agecat	1	1	0.2178	0.0818	0.0574	0.3781	7.08	0.0078
agecat	2	1	0.0396	0.0733	-0.1041	0.1833	0.29	0.5890
agecat	3	1	0.0210	0.0717	-0.1195	0.1615	0.09	0.7692
agecat	4	1	0.0032	0.0717	-0.1373	0.1437	0.00	0.9642
agecat	5	1	-0.0604	0.0784	-0.2141	0.0933	0.59	0.4412
Escal		1	1.1862	0.0123	1.1623	1.2106		

NOTE: La verosimilitud máxima ha estimado el parámetro de escala

Pure premium

Pure premium calculation

- Pure premium calculation in non life insurance is not an easy task:
  - We should combine the claim frequency with the severity.
  - Many policyholders did not have any claim (excess of zeroes).
  - The costs of claims have positive skewness.
- We have modelled the claim frequency and severity separatelly.
- If $\mathbf{N}$ is the number of claims and $\mathbf{X_i}$ is the cost of the i-th claim $(\mathrm{i} = 1, \ldots, \mathrm{N})$, then, the **total cost** will be:

$$\mathrm{S} = \begin{cases} \mathrm{X}_1 + \mathrm{X}_2 + \ldots + \mathrm{X}_\mathrm{N} & \text{si } \mathrm{N} > 0 \\ 0 & \text{si } \mathrm{N} = 0 \end{cases}$$

  Assuming independence between N and $\mathrm{X_i}$: $\mathrm{E[S] = E[N]E[X] = P}$ wich is the pure premium.
- Once we have estimated the two models separatelly we can estimate the pure premium multiplying the two expected values.
- It is also possible to perform the joint modelling of frequency and severity (for example, by using the Tweedie model).

Data analytics for non life insurance (II)
GLM

Material created by PhD. Ramon Alemany

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